SEMESTER 7 · COMPUTER VISION PROJECT

CASE FILE
NIH · 27,558

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

Automated *Microscopic* Diagnosis of Malaria

NIH cell patches · Giemsa thin smears · Chittagong Medical College Hospital

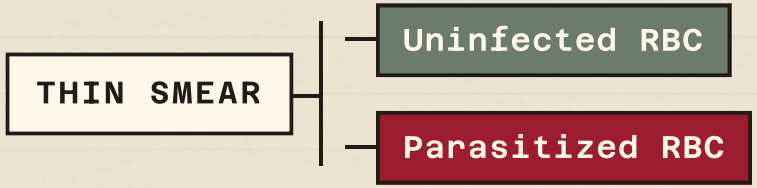
Made by — Agrapujya Lashkari, Deepanshu Mehta

LITERATURE SURVEY · 11 PAPERS · 2018–2025

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

THE PROBLEM

A person still sits at the microscope.



27,558
CELL PATCHES

13,779
EACH CLASS · BALANCED

200 pts
150 INFECTED · 50 HEALTHY

Chittagong
RELEASED BY NIH

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

Literature findings

01 Rajaraman et al., 2018 · *PeerJ* · transfer CNNs

02 Rajaraman et al., 2019 · *PeerJ* · ensemble

03 Rahman et al., 2019 · arXiv · custom CNN

04 Fuhad et al., 2020 · *Diagnostics*

05 Masud et al., 2020 · mobile CNN

06 Rajaraman et al., 2018 · MIUA · XAI

07 Madhu et al., 2023 · CapsNet · *Sci. Rep.*

08 Masud et al., 2022 · CCT transformer

09 Zedda et al., 2023 · YOLO-PAM

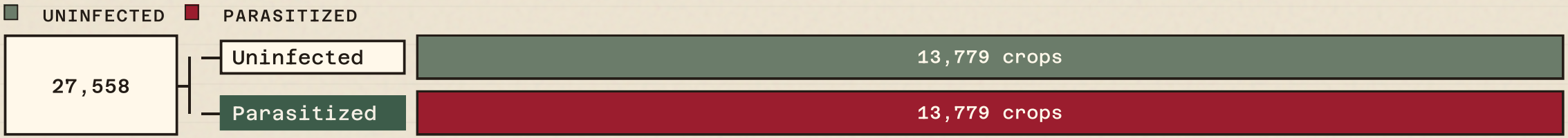
10 Mmileng et al., 2025 · ConvNeXt V2 Tiny

11 Ali et al., 2025 · DANet · *Sci. Rep.*

01 / 11 · PEERJ 2018

Rajaraman et al. — transfer CNNs

RESNET-50 / IMAGENET | FEATURE EXTRACTORS | NIH CELLS



PROBLEM

Not enough labeled smears to train a deep net from scratch. Manual features were brittle.

HOW IT WAS SOLVED

ImageNet CNNs (ResNet-50 and kin) as frozen / transferred feature extractors on cell patches.

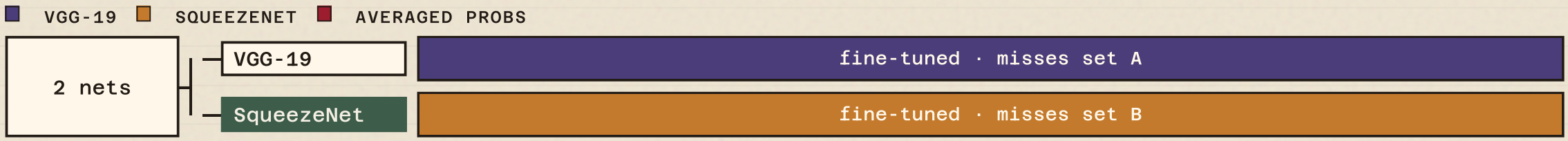
LIMIT

A single model jumped with small training-set quirks. High prediction variance.

02 / 11 · PEERJ 2019

Rajaraman et al. — ensembles

VGG-19 + SQUEEZENET | MODEL AVERAGING | COMPUTE-HEAVY



PROBLEM

One transfer model is jumpy. Unreliable for a diagnostic call.

HOW IT WAS SOLVED

Average output probabilities of two nets that fail in different places.

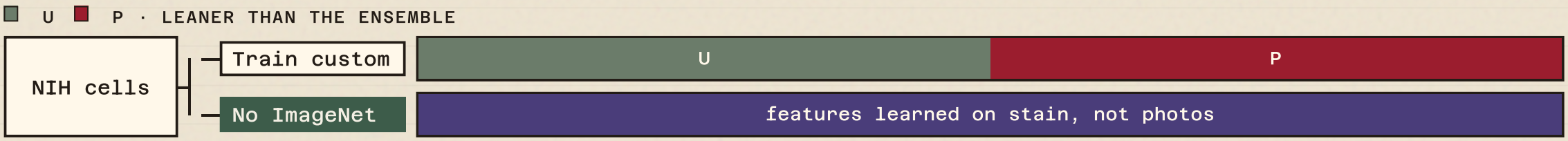
LIMIT

Too heavy for low-power clinics where point-of-care malaria testing actually happens.

03 / 11 · ARXIV 2019

Rahman et al. — custom CNN

END-TO-END CNN | FROM SCRATCH | RAW RBC PATCHES



PROBLEM

Ensembles were too bulky for field deployment.

HOW IT WAS SOLVED

A purpose-built CNN on raw red-cell patches. Skip the ImageNet stack.

LIMIT

Still a black box. Unclear if the net sees Plasmodium or stain debris.

04–05 / 11 · DIAGNOSTICS 2020 · MOBILE 2020

Fuhad · Masud — phone-scale nets

ULTRA-LIGHT CNN | SMARTPHONE / IOT | BLACK BOX

FUHAD ET AL., 2020

Deep learning on blood smears with a smartphone application. Designed for the device in the clinic, not a GPU rack.

MASUD ET AL., 2020

Mobile malaria detector. Same goal: inference that fits cheap hardware after cleaning mislabeled anomalies.

PROBLEM

Point-of-care sites cannot run VGG ensembles.

HOW IT WAS SOLVED

Tiny CNNs on cell patches. Scrub obvious label noise so the small net still trains.

LIMIT

Interpretability is gone. A doctor cannot tell parasite from dye speck.

06 / 11 · MIUA 2018

Rajaraman et al. — XAI

OCCLUSION MAPS | CAM | CLINICAL TRUST

PROBLEM

Clinicians distrust a lightweight CNN they cannot inspect.

HOW IT WAS SOLVED

Occlusion sensitivity and class activation maps show the filters tracking the parasite.

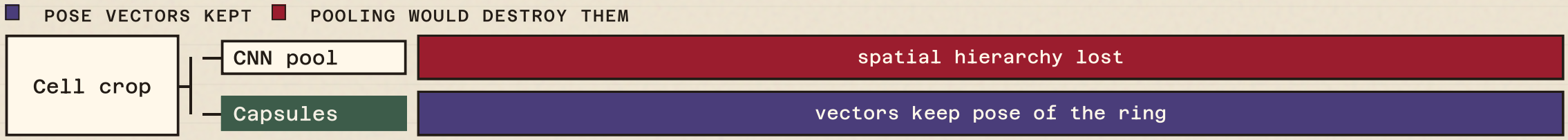
LIMIT

Max-pooling still shreds pose, shape, and orientation of the parasite.

07 / 11 · SCI. REPORTS 2023

Madhu et al. — CapsNet

INCEPTION CAPSNET | DYNAMIC ROUTING | 99.25% ACC



PROBLEM

Pooling destroys the parasite's geometry.

HOW IT WAS SOLVED

Inception-based capsules with dynamic routing. Output vectors hold spatial pose.

LIMIT

99.25% on pre-cropped cells. Math is heavy. Edge deployability undone.

08 / 11 · SENSORS 2022

Masud et al. — compact transformer

MULTI-HEAD ATTENTION | CCT | GLOBAL CELL CONTEXT

PROBLEM

CNNs see local patches. Ring-stage parasites need the whole cell as context.

HOW IT WAS SOLVED

Compact Convolutional Transformer. Self-attention over the entire cell at once.

LIMIT

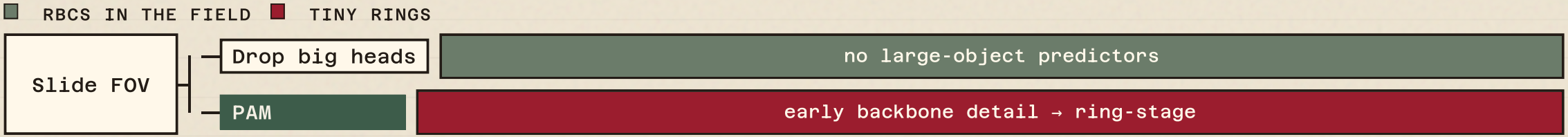
Same ceiling as CapsNet: high accuracy, high cost, still one cropped cell at a time.

> CapsNets and transformers pushed cell-level scores into the high nineties. They also walked the field back from phones toward GPUs.

09 / 11 · J. IMAGING 2023

Zedda et al. — YOLO-PAM

YOLOV8 | PARASITE ATTENTION | WHOLE FIELD OF VIEW



PROBLEM

Clinics do not crop single cells. The slide is overlapping and messy.

HOW IT WAS SOLVED

YOLOv8 minus large-object heads, plus a Parasite Attention Module on high-resolution features.

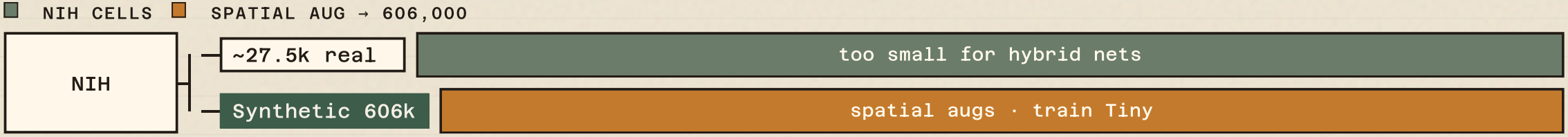
LIMIT

Fragile under data scarcity. Shear augmentations warp tiny rings until the detector misses them.

10 / 11 · IEEE ACCESS 2025

Mmileng et al. — ConvNeXt + synthetic

CONVNEXT V2 TINY | 606K SYNTHETIC | ADAMW · LABEL SMOOTHING



PROBLEM

Advanced models want huge data. Clinics want tiny compute.

HOW IT WAS SOLVED

ConvNeXt V2 Tiny (ViT hierarchy, CNN speed) on 606k synthetic images. Label smoothing + AdamW.

LIMIT

Fake data + pretrained weights. Risk of domain shift on unseen stain or strain.

11 / 11 · SCI. REPORTS 2025

Ali et al. — DANet

DILATED ATTENTION | 2.3 M PARAMS | RASPBERRY PI 4

PROBLEM

Hybrid CNN-transformers win accuracy, not clinics. Low-contrast smears blur the parasite's borders.

HOW IT WAS SOLVED

Nine dilated attention blocks (d = 1, 2, 3) widen context for free. 97.95 % accuracy with 2.3 M params.

LIMIT

Random cell-level split of one malaria dataset. Grad-CAM shown, not validated. No regularization.

DO NOT RANK ON ONE BOARD

Same NIH cells. Different exams.

#	PAPER	MOVE	UNIT	HEADLINE	COST OF THE WIN
1	Rajaraman ’18	Transfer CNN	cell	ImageNet works	high variance
2	Rajaraman ’19	Ensemble	cell	stable probs	too heavy for clinic
3	Rahman ’19	Scratch CNN	cell	end-to-end	black box
4–5	Fuhad / Masud	Tiny CNN	cell	phone / IoT	no explanation
6	Rajaraman XAI	CAM / occlusion	cell	looks at parasite	pooling kills pose
7	Madhu CapsNet	capsules	cell	99.25%	not edge-ready
8	Masud CCT	transformer	cell	global context	still one crop
9	YOLO-PAM	detection	FOV	tiny rings	aug / data fragile
10	ConvNeXt Tiny	hybrid + synth	cell	606k images	domain shift
11	DANet (Ali ’25)	dilated attention	cell	2.3 M params	cell split only

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

PHASE 1 · STILL OPEN

Four gaps before a clinic.

01 · CELL-LEVEL SPLIT LEAK

Random cell splits let the net memorize a patient's stain. Group by patient, or the nineties are an illusion.

02 · DOMAIN GENERALIZATION

NIH or synthetic clones fail on a smear from another region, dye, or strain.

03 · INSTANCE-LEVEL DETECTION

Real slides are thick and thin fields, not pre-cropped cells. Multi-instance, real-time detection is the clinical unit.

04 · CLINICALLY VALIDATED XAI

Grad-CAM is not enough. Attention must match parasitological markers: hemozoin, chromatin dots.

CASE FILE
M2 · M3 · DANET

PHASE 2 · MILESTONES 2 – 3

Reproduce DANet. Then change *one* thing.

Ali et al. · Scientific Reports 2025 · a 2.3 M-parameter dilated attention network

Made by — Agrapujya Lashkari, Deepanshu Mehta

BASELINE · REPRODUCTION GAP · HYPOTHESIS · PREDICTION BEFORE CODE

M2 · BASE PAPER · SCI. REPORTS 2025

DANet — attention on a 2.3 M budget.

9 DILATED ATTENTION BLOCKS | 2.31 M PARAMS | 0.83 GFLOPS / CELL | CODE ON GITHUB | PAPER: 97.95 % ACC

ONE DILATED ATTENTION BLOCK · AS RELEASED IN CODE

CODE AS SHIPPED

Crashes on import (NameError). The output layer has 32 classes for a 2-class task. Two small fixes.

PAPER ≠ CODE

Text: parallel dilations, sigmoid gate, LayerNorm. Code: cascaded, softmax, InstanceNorm. We ran the code: 2.31 M params matches.

OPTIMISER TRAP

Table 2's lr 0.01 never leaves 50 %. A default OneCycle call silently adds momentum. It trains with the text's 0.001, momentum 0.

M2 · REPRODUCTION · 75 EPOCHS · SEED 42 · 5,512 TEST CELLS

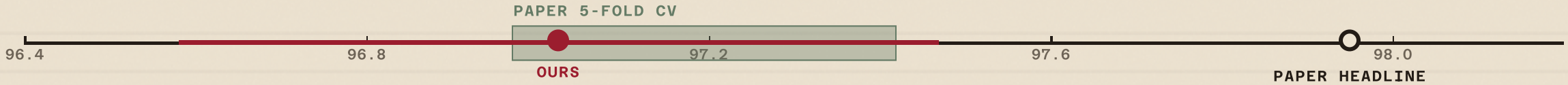
97.0 %. Inside the paper's *own* CV range.

97.02%
ACCURACY · PAPER 97.95

96.30%
SENSITIVITY · PAPER 98.07

97.75%
SPECIFICITY · PAPER 97.87

0.9945
ROC-AUC · PAPER 0.98



Accuracy, %. The red bar is our 95 % CI. The paper's headline 97.95 beats all five of its own CV folds. Our gap is almost all **sensitivity**: 102 missed infections vs 62 false alarms.

M2 · WHAT THE 97 % HIDES

Four findings before a hypothesis.

01 · EVERY TEST PATIENT WAS IN TRAINING

201 / 201 test patients also give training cells. The paper's split scores memory of a patient, not a new patient (M1 gap 01).

02 · STAIN IS A FINGERPRINT

Patient ID explains 74–90 % of a cell's mean colour. Slides differ in saturation (±48 %) and hue, but barely in brightness (±4 %).

03 · MISSES PARASITES FIRST

Sensitivity is 1.8 pts below the paper, even at matched specificity. A missed infection is the costly error.

04 · ATTENTION OFF TARGET

Grad-CAM is cold on the parasite and hot at the cell rim. It overfits after epoch ≈ 45, though the paper says it doesn't (M1 gap 04).

M3 · HYPOTHESIS · ONE VARIABLE

Hide the stain. Make it read the *parasite*.

> If we add **stain-colour augmentation** to DANet's training, **sensitivity on unseen patients will rise,** because each slide's stain is a fingerprint the network can use in place of parasite shape and chromatin.

ONE CHANGE · COLOUR JITTER	SATURATION ±50 % · HUE ±0.1	BRIGHTNESS / CONTRAST ±10 %	ALL ELSE FROZEN

MEAN CELL COLOUR · 201 PATIENTS · ONE SWATCH EACH · SORTED BY HUE

MECHANISM

Random colour on every image breaks the stain → patient → label shortcut. Only morphology stays reliable.

EVIDENCE

Patient ID explains 74–90 % of colour. The 50 healthy-only patients supply 25 % of the uninfected cells.

WHY WE MIGHT BE WRONG

The InstanceNorm in every block already strips global colour, and mean colour alone predicts the label at only 61 %.

M3 · PREDICTION · WRITTEN BEFORE ANY M4 CODE

Predict first. Then run.

ROW	CONFIG	SPLIT	ONE CHANGE	PREDICTED
B0	Baseline (M2)	cell	—	ACC 97.02 · SE 96.30 · AUC 0.9945 (measured)
B1	Baseline	patient	protocol	Balanced acc –1 to –2.5 pts · SE –1.5 to –3 pts · AUC ≈ 0.985–0.99
H	**+ colour aug**	**patient**	**colour aug**	**vs B1: SE +0.5 to +1.5 · balanced acc +0.3 to +1.0 · smaller spread across splits**
C	+ colour aug	cell	control	vs B0: within ±0.3 pts. No gain while the leak stays open

FALSIFIED IF

H ≤ B1 on sensitivity and balanced accuracy, averaged over 3 patient-level splits (40 test patients each).

REPORTED

Sensitivity, specificity, balanced accuracy, ROC-AUC, as mean ± s.d. over splits.

LIMITS

One public dataset, one hospital, no external validation. Research only, no clinical claim.